Question			Answer	Marks	AO	Guidance	
6	(a)	(i)	P (0.5, 0) Q (5.5, 0.5) R (4, 3.5) S (–1, 3)	**B1** **B1** **[2]**	**1.1** **1.1**	Any midpoint correct All midpoints correct If no labels BOD for 1 or 2 marks	

6	**(a)**	**(ii)**	Gradient PQ $= 0.1$ or length of PQ $= \frac{\sqrt{101}}{2}$ or vector $\overrightarrow{PQ} = \begin{pmatrix} 5 \\ 0.5 \end{pmatrix}$	**B1**	**1.1**	Gradient, length or vector of any one side of PQRS Or **SC1** if KLMN used	
			Gradient SR is $\frac{0.5}{5} = 0.1 =$ gradient PQ or length of SR $= \sqrt{0.5^2 + 5^2} = \frac{\sqrt{101}}{2}$ or vector $\overrightarrow{SR} = \begin{pmatrix} 4 - -1 \\ 3.5 - 3 \end{pmatrix} = \begin{pmatrix} 5 \\ 0.5 \end{pmatrix}$	**M1**	**2.2a**	Gradient of opposite side of PQRS **shown** to be equal or length or vector of opposite side **shown** to be equal. ie some working must be seen	Condone confusion of labels (eg length for gradient etc) for **M1/B1**
			Gradient PS $= -2$ or length of PS $= \frac{3\sqrt{5}}{2}$ or vector $\overrightarrow{PS} = \begin{pmatrix} -1.5 \\ 3 \end{pmatrix}$ Gradient QR is $\frac{3}{-1.5} = -2$ or length of QR $= \sqrt{1.5^2 + 3^2} = \frac{3\sqrt{5}}{2}$ or vector $\overrightarrow{QR} = \begin{pmatrix} 4 - 5.5 \\ 3.5 - 0.5 \end{pmatrix} = \begin{pmatrix} -1.5 \\ 3 \end{pmatrix}$				Watch out for valid alternatives e.g. 2 sides equal length and equal gradient
			Repeat process (as above) for other pair of opposite sides and conclude it's a parallelogram	**E1** **[3]**	**2.4**	Convincing completion dep on **M1**	

6	(b)	(i)	$\overrightarrow{WX} = -3\mathbf{c} - 3\mathbf{a} + 3\mathbf{b}$ $= 3(-\mathbf{a} + \mathbf{b} - \mathbf{c})$	**B1** **[1]**	**2.2a**	Convincing completion $\overrightarrow{WX} = \overrightarrow{WV} + \overrightarrow{VT} + \overrightarrow{TX}$	
6	(b)	(ii)	$\overrightarrow{AH} = -\mathbf{a} + \mathbf{b}$ $\overrightarrow{WX} = -3\mathbf{c} - 3\mathbf{a} + 3\mathbf{b}$ $\overrightarrow{WE} = -\mathbf{c} - \mathbf{a} + \mathbf{b}$ $\overrightarrow{DE} = \mathbf{c} - \mathbf{c} - \mathbf{a} + \mathbf{b} = -\mathbf{a} + \mathbf{b}$ So AH is parallel to DE	**B1** **E1** **[2]**	**1.1** **2.4**	$\overrightarrow{DE}$ from any correct route, must be shown Convincing completion with conclusion	
6	(b)	(iii)	$\overrightarrow{BC} = \mathbf{a} + \mathbf{c}$ $\overrightarrow{GF} = \mathbf{b} - (-\mathbf{c} - \mathbf{a} + \mathbf{b}) = \mathbf{c} + \mathbf{a}$ $\overrightarrow{BC} = \overrightarrow{GF}$ so they are parallel	**B1** **E1** **[2]**	**2.2a** **2.4**	$\overrightarrow{GF}$ from any correct route, must be shown Convincing completion with conclusion	